

Savitribai Phule Pune University Third Year of Electronics and Computer Engineering (2019 Course) 310342:Advanced Java Programming		
Teaching Scheme:	Credit: 03	Examination Scheme:
Theory: 03 Hrs. / week		In-Sem (Theory): 30 Marks End-Sem (Theory): 70 Marks
Prerequisite Courses, if any: Object Oriented Programming		
Companion Course, if any: -		
Course Objectives: 1. To design and develop basic OOPS concept in Java. 2. To design and develop packages in Java. 3. To design enterprise based applications by encapsulating an application's business logic. 4. To designing applications using pre-built frameworks.		
Course Outcomes: On completion of the course, learner will be able to- CO1: Design and develop GUI applications using Applets. CO2: Apply relevant AWT/ swing components to handle the given event. CO3: Design and develop GUI applications using Abstract Windowing Toolkit (AWT), Swing and Event Handling. CO4: Learn to access database through Java programs, using Java Database Connectivity (JDBC) CO5: Invoke the remote methods in an application using Remote Method Invocation (RMI) CO6: Develop program for client /server communication using Java Networking classes.		
Course Contents		
Unit I	OOPS Concepts	(07 Hrs.)
Inheritance Types Of Inheritance: Introduction, Classes Inheritance, Interface Inheritance, Multilevel Inheritance, Accessing members of other class/interface, Chaining Constructor using this() and Super(), This() constructor call, Super()constructor call, Dynamic method Dispatch, Method Overriding, Modifiers: Java Access Specifies: Public, Protected, Private, Default Other modifiers for members: Static, Final, Abstract, Synchronized, Native, Transient, Volatile, Other modifiers for classes: Abstract classes, Final classes.		
Mapping of Course Outcomes for Unit I	CO1: Design and develop GUI applications using Applets.	
Unit II	Packages, Interfaces and Fundamental Classes	(07 Hrs.)

String, Introduction to Packages, Types of Fundamental Classes, String Handling: Creating Format String, Exception Handling, Nested Classes, Threads, Collections and Maps, Collections, Networking: Socket Programming, URL class.		
Mapping of Course Outcomes for Unit II	CO2: Apply relevant AWT/ swing components to handle the given event.	
Unit III	Applet	(07 Hrs.)
Applet Basics -Introduction, limitations of AWT, Applet architecture – HTML APPLET tag – Passing parameter to Appletget, Japplet: Icons and Labels Text Fields Buttons, Combo Boxes, Checkboxes, Tabbed Panes, Scroll Panes.		
Mapping of Course Outcomes for Unit II	CO3: Design and develop GUI applications using Abstract Windowing Toolkit (AWT), Swing and Event Handling.	
Unit IV	Event handling using AWT/Swing components	(07 Hrs.)
Event Handling: Events, Event sources, Event classes, Event Listeners, Delegation event model, handling mouse and keyboard events, Adapter classes, inner classes. The AWT class hierarchy, user interface.		
Mapping of Course Outcomes for Unit II	CO4: Learn to access database through Java programs, using Java Database Connectivity (JDBC)	
Unit V	GUI Programming	(07 Hrs.)
Designing Graphical User Interfaces in Java, Components and Containers, Basics of Components, Using Containers, Layout Managers, AWT Components, Adding a Menu to Window, Java Utilities (java.util Package)		
Mapping of Course Outcomes for Unit II	CO5: Invoke the remote methods in an application using Remote Method Invocation (RMI)	
Unit VI	Database Programming using JDBC	(07 Hrs.)
The Concept of JDBC, JDBC Driver Types & Architecture, JDBC Packages, A Brief Overview of the JDBC process, Database Connection, Basic JDBC program Concept, Statement, Result Set, Prepared Statement, Callable Statement, Executing SQL commands, Executing queries		
Mapping of Course Outcomes for Unit VI	CO6: Develop program for client /server communication using Java Networking classes.	
Learning Resources		
Text Books: 1. Herbert Schildt, “Java: The complete reference”, Tata McGraw Hill, 7th Edition 2. Jim Keogh, “Complete Reference J2EE” , Enterpr 3. E. Balaguruswamy, “Programming with JAVA: A Primer” McGraw Hill Education, India, 5th Edition.		
Reference Books: 1. “Java 6 Programming”, Black Book, Dreamtech 2. “Java Server Programming, Java EE6 (J2EE 1.6)”, Black Book, Dreamtech		

3. M.T. Savaliya, “Advanced Java Technology”, Dreamtech

MOOC / NPTEL Courses:

1. NPTEL Course “**Programming in Java**”

Link of the Course: <https://nptel.ac.in/courses/106/105/106105191/>

2. Udemy course “**Advanced Java Programming**”

Link of the Course: <https://www.udemy.com/course/advanced-java-programming>